

## Qualification Pack



# Robotics & Automation Engineer

QP Code: CSC/Q0904

Version: 1.0

NSQF Level: 6

Capital Goods & Strategic Skill Council || 1st Floor, L-29, Outer Circle, Connaught Place  
New Delhi - 110001 || email:technicaladvisors@cgsc.in

## Qualification Pack

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## Qualification Pack

### CSC/Q0904: Robotics & Automation Engineer

#### Brief Job Description

Robotics & Automation Engineer is responsible for overseeing the design, development, testing, and implementation of robotic systems and automation solutions.

#### Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [CSC/N0914: Selection and Installation of Robotics & Automation Systems](#)
2. [CSC/N0915: Integration of Robots and Automation Systems using Industrial Networking Protocols](#)
3. [CSC/N0916: Testing & Maintenance of Robotics & Automation System.](#)
4. [CSC/N1339: Collaboratively coordinate with the team](#)
5. [CSC/N0505: Follow health, safety and environment guidelines at workplace](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                                                           |
|-------------------|-----------------------------------------------------------|
| <b>Sector</b>     | Capital Goods                                             |
| <b>Sub-Sector</b> | Robotics and Automation, Smart Manufacturing, Maintenance |
| <b>Occupation</b> | Maintenance                                               |
| <b>Country</b>    | India                                                     |
| <b>NSQF Level</b> | 6                                                         |
| <b>Credits</b>    | 22                                                        |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/7412.0101                                                                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Educational Qualification &amp; Experience</b> | Completed 4 year UG degree with Honours/ Honours with research (or equivalent)<br>OR<br>Completed 2nd year diploma after 12th with 3 Years of experience relevant<br>OR<br>Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience relevant<br>OR<br>Previous relevant Qualification of NSQF Level (5) with 3 Years of experience relevant |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                            |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Next Review Date</b>                                   | 08/05/2028                                                                                                                                                                                                                                                                                                                                                          |
| <b>NSQC Approval Date</b>                                 | 08/05/2025                                                                                                                                                                                                                                                                                                                                                          |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Reference code on NQR</b>                              | QG-06-IT-04197-2025-V1-CGSC                                                                                                                                                                                                                                                                                                                                         |
| <b>NQR Version</b>                                        | 1.0                                                                                                                                                                                                                                                                                                                                                                 |

## Qualification Pack

# CSC/N0914: Selection and Installation of Robotics & Automation Systems

## Description

This NOS unit is about Selection and installation of Robotics & Automation Systems.

## Scope

The scope covers the following :

- Select & Develop Robotics & Automation Systems layout and Component drawings.
- Perform installation of Robotics & Automation system.
- Perform post-installation activities.

## Elements and Performance Criteria

### *Select & Develop Robotics & Automation Develop Systems layout and Component drawings.*

To be competent, the user/individual on the job must be able to:

- PC1.** identify and select the automation elements in align with electrical, mechanical and environmental parameters like power, response time, inbuilt protections, dimension, weight, package, thermal characteristics etc. as per the e-plan and required specifications.
- PC2.** prepare / interpret the project design by obtaining information from mechanical drawings and layout diagram.
- PC3.** Identify and mitigate any potential risks associated with the design, such as safety hazards, performance bottlenecks, or compatibility issues.

### *Perform installation of Robotics & Automation system.*

To be competent, the user/individual on the job must be able to:

- PC4.** Perform assembly of the system components like D.C. valve, cylinder assembly, Sensor Mounting etc. as per the mechanical drawings and SOP.
- PC5.** Follow the per-commissioning to ensure proper functionality of installed Robotics & automation Systems.
- PC6.** Carry out the various integration activities like programming, wiring with external elements etc. as per design document and SOP.

### *Perform post-installation activities.*

To be competent, the user/individual on the job must be able to:

- PC7.** conduct the trials of Robotics & automation system as per the e-plan to align it with existing or new manufacturing process.
- PC8.** handover the system to production team & train them on it as per organizational guidelines and procedures.
- PC9.** prepare documents and records such as experience under development, TGW /TGR faced during process trials etc. as a reference for future development
- PC10.** perform dry-run of the automation system with the existing manufacturing process.

## Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** Understanding of Robotics and Automation Technologies: A deep understanding of various types of robotics systems (e.g., industrial robots, collaborative robots, autonomous mobile robots) and automation technologies (e.g., PLCs, SCADA systems, HMI interfaces) is crucial. This includes knowledge of their capabilities, limitations, and applications across different industries.
- KU4.** Mechanical Engineering: Proficiency in mechanical engineering principles is essential for selecting and installing robotics systems. This includes knowledge of mechanical design, kinematics, dynamics, and materials science to ensure proper integration and functionality within the intended environment.
- KU5.** Electrical Engineering: Understanding electrical systems and components is necessary for integrating robotics and automation solutions. This includes knowledge of electrical circuits, power distribution, motors, sensors, and actuators, as well as proficiency in reading electrical schematics and diagrams.
- KU6.** Programming and Software Development: Proficiency in programming languages such as Python, C plus plus, or Java is important for configuring, programming, and controlling robotic systems. Additionally, familiarity with robot operating systems (ROS) and automation software platforms is beneficial for customizing and optimizing system performance.
- KU7.** Control Systems Engineering: Knowledge of control systems theory and techniques is essential for designing and implementing closed-loop control algorithms to regulate the behavior and performance of robotic systems. This includes understanding PID control, motion control, trajectory planning, and feedback mechanisms.
- KU8.** Safety Standards and Regulations: Understanding safety standards and regulations relevant to robotics and automation systems (e.g., ISO 10218, ISO 13849, ANSI/RIA R15.06) is crucial for ensuring the safety of personnel and equipment during installation and operation. This includes knowledge of risk assessment methodologies and safety system design principles.
- KU9.** Project Management: Proficiency in project management principles and practices is necessary for planning, coordinating, and executing robotics and automation projects effectively. This includes skills in budgeting, scheduling, resource allocation, and risk management to ensure projects are completed on time and within budget.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks.
- GS4.** complete tasks efficiently and accurately within stipulated time.
- GS5.** inform/report to concerned person in case of any problem.
- GS6.** make timely decisions for efficient utilization of resources.



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**GS7.** write reports such as accident report, in at least English/regional language



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Select &amp; Develop Robotics &amp; Automation Develop Systems layout and Component drawings.</i>                                                                                                                                                                              | <b>10</b>    | <b>10</b>       | -             | <b>6</b>   |
| <b>PC1.</b> identify and select the automation elements in align with electrical, mechanical and environmental parameters like power, response time, inbuilt protections, dimension, weight, package, thermal characteristics etc. as per the e-plan and required specifications. | 3            | 2               | -             | 2          |
| <b>PC2.</b> prepare / interpret the project design by obtaining information from mechanical drawings and layout diagram.                                                                                                                                                          | 3            | 4               | -             | 2          |
| <b>PC3.</b> Identify and mitigate any potential risks associated with the design, such as safety hazards, performance bottlenecks, or compatibility issues.                                                                                                                       | 4            | 4               | -             | 2          |
| <i>Perform installation of Robotics &amp; Automation system.</i>                                                                                                                                                                                                                  | <b>15</b>    | <b>15</b>       | -             | <b>10</b>  |
| <b>PC4.</b> Perform assembly of the system components like D.C. valve, cylinder assembly, Sensor Mounting etc. as per the mechanical drawings and SOP.                                                                                                                            | 5            | 5               | -             | 3          |
| <b>PC5.</b> Follow the per-commissioning to ensure proper functionality of installed Robotics & automation Systems.                                                                                                                                                               | 5            | 5               | -             | 3          |
| <b>PC6.</b> Carry out the various integration activities like programming, wiring with external elements etc. as per design document and SOP.                                                                                                                                     | 5            | 5               | -             | 4          |
| <i>Perform post-installation activities.</i>                                                                                                                                                                                                                                      | <b>15</b>    | <b>15</b>       | -             | <b>4</b>   |
| <b>PC7.</b> conduct the trials of Robotics & automation system as per the e-plan to align it with existing or new manufacturing process.                                                                                                                                          | 5            | 4               | -             | 1          |
| <b>PC8.</b> handover the system to production team & train them on it as per organizational guidelines and procedures.                                                                                                                                                            | 4            | 3               | -             | 1          |



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| Assessment Criteria for Outcomes                                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC9.</b> prepare documents and records such as experience under development, TGW /TGR faced during process trials etc. as a reference for future development | 3            | 4               | -             | 1          |
| <b>PC10.</b> perform dry-run of the automation system with the existing manufacturing process.                                                                  | 3            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |

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### National Occupational Standards (NOS) Parameters

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0914                                                   |
| <b>NOS Name</b>            | Selection and Installation of Robotics & Automation Systems |
| <b>Sector</b>              | Capital Goods                                               |
| <b>Sub-Sector</b>          |                                                             |
| <b>Occupation</b>          | Maintenance                                                 |
| <b>NSQF Level</b>          | 6                                                           |
| <b>Credits</b>             | 3                                                           |
| <b>Version</b>             | 1.0                                                         |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                  |
| <b>Next Review Date</b>    | 08/05/2028                                                  |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                  |

## Qualification Pack

# CSC/N0915: Integration of Robots and Automation Systems using Industrial Networking Protocols

## Description

This unit is about to perform Integration of Robots and Automation Systems using Industrial Networking Protocols

## Scope

The scope covers the following :

- Perform installation and commissioning of robot & Automation System
- Select the Suitable Industrial Networking Protocols to Establish Communication
- Perform post-System integration activities

## Elements and Performance Criteria

### *Perform installation and commissioning of robot & Automation System*

To be competent, the user/individual on the job must be able to:

- PC1.** Perform the per-commissioning to ensure proper functionality of safety elements like operator station controls (E-stop, cycle start, reset), safety peripherals such as light curtains, gate plug connections, tooling and teach pendant controls
- PC2.** Turn on the power of robot & Automation System, look for any warnings / errors in it and rectify the same as per as per organizational guidelines
- PC3.** Set the initial settings like robot jogging, mastering, axis limits, PLC Parameters, Hard wiring in a standalone environment as per guidelines along with robot & PLC manual.

### *Select the Suitable Industrial Networking Protocols to Establish Communication*

To be competent, the user/individual on the job must be able to:

- PC4.** Analyse the installed Machines, automation elements, system and robots into different layers of network architecture like field devices, control devices ,network.
- PC5.** Design / interpret the network consists of devices, automation system and robots, connect the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document
- PC6.** select and install the suitable network protocols like MODBUS,CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. based on control system requirements.

### *Perform post-System integration activities.*

To be competent, the user/individual on the job must be able to:

- PC7.** perform the backup processes like teach pendant programs, PLC & Robot parameter, mastering data using different backup devices as per organizational guidelines.
- PC8.** handover the system to production team & train them on it as per organizational guidelines and procedures.
- PC9.** prepare documents and records such as experience under development, TGW /TGR faced during process trials etc. as a reference for future development.

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- PC10.** identify critical spares like encoder, encoder battery, dedicated fuse , PLC Backup Battery,Communication Cable etc. with the help of supplier, maintenance team and plan for their availability

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** Industrial Networking Protocols: In-depth understanding of industrial communication protocols such as Ethernet/IP, PROFINET, Modbus TCP, EtherCAT, and DeviceNet. Knowledge of the OSI model and TCP/IP stack is essential for troubleshooting and optimizing network performance.
- KU4.** Robotics Systems: Familiarity with robotic hardware, software, and programming languages (e.g., ROS, PLC programming). Understanding robot kinematics, motion control, and end-of-arm tooling is crucial for integrating robots into automation systems.
- KU5.** Automation Systems: Knowledge of automation components such as programmable logic controllers (PLCs), human-machine interfaces (HMIs), sensors, actuators, and motion controllers. Understanding industrial automation standards (e.g., IEC 61131) and control system architectures is essential.
- KU6.** System Integration: Ability to design, configure, and integrate robotic and automation systems into existing manufacturing processes or production lines. Proficiency in system integration tools and techniques, including network configuration, device configuration, and data mapping.
- KU7.** Network Infrastructure: Understanding of industrial network infrastructure components, including switches, routers, gateways, and firewalls. Knowledge of network segmentation, VLANs, and subnetting to ensure secure and efficient communication between devices.
- KU8.** Troubleshooting and Diagnostics: Strong troubleshooting skills to identify and resolve network issues, communication errors, and interoperability problems. Proficiency in network monitoring tools and diagnostic utilities for analyzing network traffic and performance metrics.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations



## Qualification Pack

**GS9.** check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Perform installation and commissioning of robot &amp; Automation System</i>                                                                                                                                                                                | <b>10</b>    | <b>10</b>       | -             | <b>6</b>   |
| <b>PC1.</b> Perform the per-commissioning to ensure proper functionality of safety elements like operator station controls (E-stop, cycle start, reset), safety peripherals such as light curtains, gate plug connections, tooling and teach pendant controls | 3            | 2               | -             | 2          |
| <b>PC2.</b> Turn on the power of robot & Automation System, look for any warnings / errors in it and rectify the same as per as per organizational guidelines                                                                                                 | 3            | 4               | -             | 2          |
| <b>PC3.</b> Set the initial settings like robot jogging, mastering, axis limits, PLC Parameters, Hard wiring in a standalone environment as per guidelines along with robot & PLC manual.                                                                     | 4            | 4               | -             | 2          |
| <i>Select the Suitable Industrial Networking Protocols to Establish Communication</i>                                                                                                                                                                         | <b>15</b>    | <b>15</b>       | -             | <b>10</b>  |
| <b>PC4.</b> Analyse the installed Machines, automation elements, system and robots into different layers of network architecture like field devices, control devices ,network.                                                                                | 5            | 5               | -             | 3          |
| <b>PC5.</b> Design / interpret the network consists of devices, automation system and robots, connect the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document                                 | 5            | 5               | -             | 3          |
| <b>PC6.</b> select and install the suitable network protocols like MODBUS,CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. based on control system requirements.                                                                                                | 5            | 5               | -             | 4          |
| <i>Perform post-System integration activities.</i>                                                                                                                                                                                                            | <b>15</b>    | <b>15</b>       | -             | <b>4</b>   |
| <b>PC7.</b> perform the backup processes like teach pendant programs, PLC & Robot parameter, mastering data using different backup devices as per organizational guidelines.                                                                                  | 5            | 5               | -             | 1          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC8.</b> handover the system to production team & train them on it as per organizational guidelines and procedures.                                                                                        | 3            | 3               | -             | 1          |
| <b>PC9.</b> prepare documents and records such as experience under development, TGW /TGR faced during process trials etc. as a reference for future development.                                              | 3            | 3               | -             | 1          |
| <b>PC10.</b> identify critical spares like encoder, encoder battery, dedicated fuse , PLC Backup Battery,Communication Cable etc. with the help of supplier, maintenance team and plan for their availability | 4            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                                              | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                    |
|----------------------------|------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0915                                                                          |
| <b>NOS Name</b>            | Integration of Robots and Automation Systems using Industrial Networking Protocols |
| <b>Sector</b>              | Capital Goods                                                                      |
| <b>Sub-Sector</b>          |                                                                                    |
| <b>Occupation</b>          | Maintenance                                                                        |
| <b>NSQF Level</b>          | 6                                                                                  |
| <b>Credits</b>             | 6                                                                                  |
| <b>Version</b>             | 1.0                                                                                |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                                         |
| <b>Next Review Date</b>    | 08/05/2028                                                                         |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                                         |

## Qualification Pack

### CSC/N0916: Testing & Maintenance of Robotics & Automation System.

#### Description

This unit is about to Perform Testing & Maintenance of Robotics & Automation System.

#### Scope

The scope covers the following :

- Fetching vital data from robotics and automation system using industrial networking.
- Testing of Robotics & Automation System using Simulators & Dry Run.
- Carry out maintenance and troubleshooting of communication network between robotics and automation system.

#### Elements and Performance Criteria

##### *Fetching vital data from robotics and automation system using industrial networking.*

To be competent, the user/individual on the job must be able to:

- PC1.** Analyse the installed automation elements, system and robots into different layers of network architecture like field devices, control devices ,network.
- PC2.** Fetch the machine data of robotics and automation system like communication status, healthy hardware, life diagnosis, condition monitoring as per organizational guidelines
- PC3.** Fetch the process and control data of robotics and automation system like cycle start/stop, production status, operator status, system monitor as per organizational guidelines.

##### *Testing of Robotics & Automation System using Simulators & Dry Run.*

To be competent, the user/individual on the job must be able to:

- PC4.** Verify that the robotics and automation systems perform their intended functions accurately and efficiently.
- PC5.** Evaluate the reliability of the system under different operating conditions and stress levels. Robustness testing involves subjecting the system to variations in input parameters, environmental conditions, and unexpected events to ensure stable
- PC6.** Test the system's ability to detect and recover from errors and faults autonomously or with minimal human intervention. This includes simulating various failure scenarios to assess the system's resilience.

##### *Carry out maintenance and troubleshooting of communication network between robotics and automation system.*

To be competent, the user/individual on the job must be able to:

- PC7.** prepare maintenance and troubleshooting schedule and checklist for conducting the preventive, predictive and breakdown maintenance of the industrial networking
- PC8.** Identify the critical spares with the help of supplier, maintenance team and plan for their availability
- PC9.** develop the maintenance manual with the help of supplier and maintenance team.

#### Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** Mechanical Systems: Understanding of mechanical components and systems such as motors, gears, actuators, and manipulators. Knowledge of mechanical principles, including kinematics and dynamics, is essential for troubleshooting and maintaining robotic systems.
- KU4.** Electrical and Electronics: Proficiency in electrical and electronic systems, including sensors, controllers, power systems, and communication interfaces. Understanding electrical circuits, signal processing, and control systems is crucial for testing and diagnosing issues in automation systems.
- KU5.** Programming and Software: Strong programming skills in languages commonly used in robotics and automation, such as Python, C/C++ or MATLAB. Knowledge of robotic frameworks and middleware (e.g., ROS - Robot Operating System) is valuable for developing and testing control algorithms and software applications.
- KU6.** Sensors and Perception: Familiarity with various types of sensors used in robotics, such as proximity sensors, vision systems, LiDAR, and inertial sensors. Understanding sensor fusion techniques and perception algorithms is essential for testing and validating sensor data processing and interpretation.
- KU7.** Control Systems: Knowledge of control theory and techniques for feedback control, motion planning, trajectory generation, and path following. Understanding PID controllers, state-space control, and optimization algorithms is important for testing and tuning control algorithms in robotic systems.
- KU8.** Safety Standards and Regulations: Awareness of safety standards and regulations applicable to robotics and automation systems, such as ISO 10218 (Safety of Industrial Robots) and ISO 13849 (Safety of Machinery). Understanding safety protocols and risk assessment methodologies is crucial for ensuring safe operation and maintenance practices.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Fetching vital data from robotics and automation system using industrial networking.</i>                                                                                                                                                                      | 10           | 10              | -             | 7          |
| <b>PC1.</b> Analyse the installed automation elements, system and robots into different layers of network architecture like field devices, control devices ,network.                                                                                             | 3            | 3               | -             | 2          |
| <b>PC2.</b> Fetch the machine data of robotics and automation system like communication status, healthy hardware, life diagnosis, condition monitoring as per organizational guidelines                                                                          | 3            | 3               | -             | 2          |
| <b>PC3.</b> Fetch the process and control data of robotics and automation system like cycle start/stop, production status, operator status, system monitor as per organizational guidelines.                                                                     | 4            | 4               | -             | 3          |
| <i>Testing of Robotics &amp; Automation System using Simulators &amp; Dry Run.</i>                                                                                                                                                                               | 15           | 15              | -             | 9          |
| <b>PC4.</b> Verify that the robotics and automation systems perform their intended functions accurately and efficiently.                                                                                                                                         | 5            | 5               | -             | 3          |
| <b>PC5.</b> Evaluate the reliability of the system under different operating conditions and stress levels. Robustness testing involves subjecting the system to variations in input parameters, environmental conditions, and unexpected events to ensure stable | 5            | 5               | -             | 3          |
| <b>PC6.</b> Test the system's ability to detect and recover from errors and faults autonomously or with minimal human intervention. This includes simulating various failure scenarios to assess the system's resilience.                                        | 5            | 5               | -             | 3          |
| <i>Carry out maintenance and troubleshooting of communication network between robotics and automation system.</i>                                                                                                                                                | 15           | 15              | -             | 4          |
| <b>PC7.</b> prepare maintenance and troubleshooting schedule and checklist for conducting the preventive, predictive and breakdown maintenance of the industrial networking                                                                                      | 5            | 5               | -             | 2          |

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| Assessment Criteria for Outcomes                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC8.</b> Identify the critical spares with the help of supplier, maintenance team and plan for their availability | 5            | 5               | -             | 1          |
| <b>PC9.</b> develop the maintenance manual with the help of supplier and maintenance team.                           | 5            | 5               | -             | 1          |
| <b>NOS Total</b>                                                                                                     | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0916                                              |
| <b>NOS Name</b>            | Testing & Maintenance of Robotics & Automation System. |
| <b>Sector</b>              | Capital Goods                                          |
| <b>Sub-Sector</b>          |                                                        |
| <b>Occupation</b>          | Maintenance                                            |
| <b>NSQF Level</b>          | 6                                                      |
| <b>Credits</b>             | 7                                                      |
| <b>Version</b>             | 1.0                                                    |
| <b>Last Reviewed Date</b>  | 08/05/2025                                             |
| <b>Next Review Date</b>    | 08/05/2028                                             |
| <b>NSQC Clearance Date</b> | 08/05/2025                                             |

## Qualification Pack

### CSC/N1339: Collaboratively coordinate with the team

#### Description

This OS unit is about building relationships and working with people and groups inside and outside the organization, using skills and habits, to achieve the team goals and objectives

#### Scope

The scope covers the following :

- This unit/task covers the following:
- Creating team environment
- Communicating - giving and receiving
- Working cooperatively
- Participating in team decision making
- Demonstrating Sense of Responsibility
- Showing respect for opinions, customs, and preferences

#### Elements and Performance Criteria

##### *Communicate effectively at the workplace*

To be competent, the user/individual on the job must be able to:

- PC1.** exchange information and instruction with colleagues, and seek clarifications and feedback
- PC2.** assist colleagues where required
- PC3.** follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)
- PC4.** document and share all relevant information with stakeholders in agreed formats and as per agreed timelines

##### *Work effectively*

To be competent, the user/individual on the job must be able to:

- PC5.** identify and obtain clarity regarding organisational, team and own goals and targets
- PC6.** prioritise and plan work in order to achieve goals and targets
- PC7.** monitor own and team performance as per agreed plan
- PC8.** complete duties accurately, systematically and within required timeframes
- PC9.** express emotions appropriately at the workplace and manage own response to heightened emotions
- PC10.** maintain orderliness and cleanliness in the work area Maintain and enhance professional competence
- PC11.** identify own strengths and weaknesses in relation to goals and targets
- PC12.** adapt self, service, or product to meet success criteria
- PC13.** seek and select opportunities for continuous professional development
- PC14.** formulate a professional development plan to enhance capabilities

## Qualification Pack

- PC15.** build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations
- PC16.** examine developments and trends in field of work and their potential impact on work
- PC17.** take feedback from peers, supervisors and clients to improve own performance and practices  
*Work in a disciplined and ethical manner*

To be competent, the user/individual on the job must be able to:

- PC18.** perform tasks as per workplace standards, organizational policies and legislative requirements
- PC19.** display appropriate professional appearance at the workplace and adhere to the organizational dress code
- PC20.** demonstrate responsible and disciplined behavior at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behavior at all times, adopting environment- friendly practices, etc.
- PC21.** identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution
- PC22.** protect the rights of the client and organization when delivering services
- PC23.** ensure services are delivered equally to all clients regardless of personal and cultural beliefs
- PC24.** operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities
- PC25.** follow organizational guidelines and legal requirements on disclosure and confidentiality

*Uphold social diversity at the workplace*

To be competent, the user/individual on the job must be able to:

- PC26.** recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes
- PC27.** identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace
- PC28.** use inclusive or neutral language and gestures in all interactions
- PC29.** respect the personal and professional space of others
- PC30.** access grievance redressal mechanisms as per legislations

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organisation's policies and procedures for working with colleagues, roles and responsibilities
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** different methods of communication and the circumstances in which it is appropriate to use these
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** different types of information that colleagues might need and the importance of providing this information when it is required



## Qualification Pack

- KU7.** the importance of helping colleagues with problems, to meet quality and time standards as a team

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write instructions, guidelines, procedures, messages, emails, and other media in language of the workplace
- GS2.** communicate in common and technical terms in language of the workplace
- GS3.** listen effectively and orally communicate information
- GS4.** be punctual, do work scheduling and reporting
- GS5.** comply with workplace practices and ethics
- GS6.** maintain cleanliness and healthy environment
- GS7.** be customer friendly - understand real needs of the customer and suggest most appropriate solution
- GS8.** be safety conscious and avoid risk
- GS9.** be observant, vigilant, and security consciousness
- GS10.** respond, handle problem, and escalate as necessary
- GS11.** ask for clarification and advice from concerned persons
- GS12.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS13.** plan and organize work to achieve targets and deadlines

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Communicate effectively at the workplace</i>                                                                                    | <b>7</b>     | <b>20</b>       | -             | -          |
| <b>PC1.</b> exchange information and instruction with colleagues, and seek clarifications and feedback                             | -            | -               | -             | -          |
| <b>PC2.</b> assist colleagues where required                                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person) | -            | -               | -             | -          |
| <b>PC4.</b> document and share all relevant information with stakeholders in agreed formats and as per agreed timelines            | -            | -               | -             | -          |
| <i>Work effectively</i>                                                                                                            | <b>7</b>     | <b>20</b>       | -             | -          |
| <b>PC5.</b> identify and obtain clarity regarding organisational, team and own goals and targets                                   | -            | -               | -             | -          |
| <b>PC6.</b> prioritise and plan work in order to achieve goals and targets                                                         | -            | -               | -             | -          |
| <b>PC7.</b> monitor own and team performance as per agreed plan                                                                    | -            | -               | -             | -          |
| <b>PC8.</b> complete duties accurately, systematically and within required timeframes                                              | -            | -               | -             | -          |
| <b>PC9.</b> express emotions appropriately at the workplace and manage own response to heightened emotions                         | -            | -               | -             | -          |
| <b>PC10.</b> maintain orderliness and cleanliness in the work area Maintain and enhance professional competence                    | -            | -               | -             | -          |
| <b>PC11.</b> identify own strengths and weaknesses in relation to goals and targets                                                | -            | -               | -             | -          |
| <b>PC12.</b> adapt self, service, or product to meet success criteria                                                              | -            | -               | -             | -          |
| <b>PC13.</b> seek and select opportunities for continuous professional development                                                 | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> formulate a professional development plan to enhance capabilities                                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC15.</b> build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations                                                                                                                            | -            | -               | -             | -          |
| <b>PC16.</b> examine developments and trends in field of work and their potential impact on work                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC17.</b> take feedback from peers, supervisors and clients to improve own performance and practices                                                                                                                                                | -            | -               | -             | -          |
| <i>Work in a disciplined and ethical manner</i>                                                                                                                                                                                                        | <b>8</b>     | <b>20</b>       | -             | -          |
| <b>PC18.</b> perform tasks as per workplace standards, organizational policies and legislative requirements                                                                                                                                            | -            | -               | -             | -          |
| <b>PC19.</b> display appropriate professional appearance at the workplace and adhere to the organizational dress code                                                                                                                                  | -            | -               | -             | -          |
| <b>PC20.</b> demonstrate responsible and disciplined behavior at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behavior at all times, adopting environment- friendly practices, etc. | -            | -               | -             | -          |
| <b>PC21.</b> identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution                                                                | -            | -               | -             | -          |
| <b>PC22.</b> protect the rights of the client and organization when delivering services                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC23.</b> ensure services are delivered equally to all clients regardless of personal and cultural beliefs                                                                                                                                          | -            | -               | -             | -          |
| <b>PC24.</b> operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities                                                                                                                             | -            | -               | -             | -          |
| <b>PC25.</b> follow organizational guidelines and legal requirements on disclosure and confidentiality                                                                                                                                                 | -            | -               | -             | -          |
| <i>Uphold social diversity at the workplace</i>                                                                                                                                                                                                        | <b>8</b>     | <b>10</b>       | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC26.</b> recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes | -            | -               | -             | -          |
| <b>PC27.</b> identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace                               | -            | -               | -             | -          |
| <b>PC28.</b> use inclusive or neutral language and gestures in all interactions                                                                                   | -            | -               | -             | -          |
| <b>PC29.</b> respect the personal and professional space of others                                                                                                | -            | -               | -             | -          |
| <b>PC30.</b> access grievance redressal mechanisms as per legislations                                                                                            | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                  | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                          |
|----------------------------|------------------------------------------|
| <b>NOS Code</b>            | CSC/N1339                                |
| <b>NOS Name</b>            | Collaboratively coordinate with the team |
| <b>Sector</b>              | Capital Goods                            |
| <b>Sub-Sector</b>          | Generic                                  |
| <b>Occupation</b>          | Generic                                  |
| <b>NSQF Level</b>          | 5                                        |
| <b>Credits</b>             | 3                                        |
| <b>Version</b>             | 1.0                                      |
| <b>Last Reviewed Date</b>  | 01/10/2025                               |
| <b>Next Review Date</b>    | 01/10/2030                               |
| <b>NSQC Clearance Date</b> | 01/10/2025                               |

## Qualification Pack

### CSC/N0505: Follow health, safety and environment guidelines at workplace

#### Description

This OS unit is about following adequate safety procedures to make work environment healthy and safe

#### Scope

The scope covers the following :

- This unit/task covers the following:
- Adhere to standard safety procedures of the company
- Follow healthy practices and posture
- Practice waste management and recycling
- Conserve material and resources

#### Elements and Performance Criteria

##### *Adhere to standard safety procedures of the organisation*

To be competent, the user/individual on the job must be able to:

- PC1.** comply with general safety procedures and those for handling equipment, tools, chemicals, and hazardous material, as prescribed and followed in the organisation
- PC2.** remove finger rings or any other metal objects likely to interfere with the work
- PC3.** ensure that identification badge or any other object worn around the neck or on the clothing does not get caught in any rotating machine, or otherwise interfere with the work
- PC4.** use appropriate safety devices such as goggles, gloves, ear plugs, caps, ESD pins, covers, shoes, helmets etc. recommended for the work being performed
- PC5.** inform, escalate, or raise alarm about any suspicions, unaccounted hazardous material, devices, or other objects found in the premises
- PC6.** inform, escalate, or raise alarm about any breach of safety or security procedure in the organisation
- PC7.** help achieve zero accidents goals at work
- PC8.** avoid damage to sensitive electronic components due to negligence of ESD procedures
- PC9.** participate regularly in fire drills or other safety related workshops organised by the organisation
- PC10.** follow strictly all access control and perimeter safety procedures in designated factory areas such as robotic work stations, automated production lines, automated material movement and other potentially risky operations
- PC11.** ensure that other people follow all access control and perimeter safety procedures in designated factory areas and help avoid accidents
- PC12.** use emergency switches or other mechanisms of stopping a machine immediately in case any emergency situation has developed or about to happen
- PC13.** ensure that electrical equipment are properly grounded
- PC14.** follow Cyber Security guidelines and be vigilant at workplace

## Qualification Pack

**PC15.** proceed to designated safe assembly area immediately on hearing fire alarm

*Follow healthy practices and posture*

To be competent, the user/individual on the job must be able to:

**PC16.** wash hands and use sanitizers as recommended to prevent spread of diseases

**PC17.** follow common personal hygiene practices

**PC18.** maintain appropriate posture, especially in long hours of sitting or standing position and in handling heavy materials

**PC19.** participate in company organised health sessions such as exercises, games, yoga, physiotherapy, and other activities

**PC20.** handle heavy and hazardous materials with care, while maintaining appropriate posture, using suitable tools, and handling equipment such as trolleys, jacks, and ladders

**PC21.** learn and apply first aid devices available in the workplace

**PC22.** learn and apply safety and handling procedures for electrical shock and electrocution

**PC23.** learn and apply emergency medical help services

**PC24.** follow workplace decorum and avoid emotional outbursts or inappropriate language

**PC25.** prevent any harassment at workplace

*Practice waste management and recycling*

To be competent, the user/individual on the job must be able to:

**PC26.** identify recyclable, non-recyclable, and hazardous waste generated in the workplace and comply with their disposal procedures

**PC27.** dispose non-recyclable waste and hazardous waste following recommended processes

**PC28.** deposit recyclable and reusable material at identified locations

**PC29.** support education and compliance of waste management processes

*Conserve material and resources*

To be competent, the user/individual on the job must be able to:

**PC30.** identify ways to optimize usage of material and resources such as water, electricity, energy in various tasks, activities, and processes

**PC31.** check for spills and leakages of material in various tasks, activities, and processes and plug them

**PC32.** escalate the leakage issue to appropriate authority if needed

**PC33.** carry out routine cleaning of tools, machines, and equipment and maintain them in good working condition to optimize efficiency and wastage

**PC34.** check if the equipment is functioning normally before commencing work and rectify or report any malfunctioning to the responsible agency

**PC35.** check for any odour, sparks, fumes, emission, unusual vibration, noise, or any other objectionable presence in the environment and take immediate corrective action followed by report to responsible agency

**PC36.** ensure electrical equipment are properly connected for use and are switched off when not in use

**PC37.** support education and compliance of resource conservation processes

## Knowledge and Understanding (KU)

## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** company policies on workplace, environment, and personnel management
- KU2.** company policy on occupational safety and health
- KU3.** professional hazards related to nature of work and how to deal with them
- KU4.** how to maintain the work area safe and secure
- KU5.** how to handle hazardous materials, tools, and equipment
- KU6.** emergency procedures for fire, electrocution, physical injury, wounds, etc.
- KU7.** need for proper body posture and use of appropriate handling equipment
- KU8.** understand electrical grounding practices
- KU9.** common sources of pollution and ways to minimize it
- KU10.** waste management - categorisation, colour coding, handling, and disposal procedure
- KU11.** organisation policies and procedures for minimizing waste
- KU12.** efficient use of electricity, material, and water in processes
- KU13.** organization policies regarding network usage and security
- KU14.** norms for professional behaviour at workplace and dealing with deviations

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicating in the language of the workplace
- GS2.** reading and interpreting documents, drawings, symbols, and instructions
- GS3.** operating computer and common office equipment and diagnosing common electrical and interconnection problems
- GS4.** writing notes, reports, observations, emails
- GS5.** using personnel protective devices
- GS6.** maintaining clean and healthy work environment
- GS7.** using and operating safety devices and equipment
- GS8.** conducting work following workplace security processes and rules
- GS9.** responding to emergency situations pertaining to workplace
- GS10.** understanding people and collaborating to create a healthy workplace



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Adhere to standard safety procedures of the organisation</i>                                                                                                                                                                           | <b>7</b>     | <b>10</b>       | -             | -          |
| <b>PC1.</b> comply with general safety procedures and those for handling equipment, tools, chemicals, and hazardous material, as prescribed and followed in the organisation                                                              | -            | -               | -             | -          |
| <b>PC2.</b> remove finger rings or any other metal objects likely to interfere with the work                                                                                                                                              | -            | -               | -             | -          |
| <b>PC3.</b> ensure that identification badge or any other object worn around the neck or on the clothing does not get caught in any rotating machine, or otherwise interfere with the work                                                | -            | -               | -             | -          |
| <b>PC4.</b> use appropriate safety devices such as goggles, gloves, ear plugs, caps, ESD pins, covers, shoes, helmets etc. recommended for the work being performed                                                                       | -            | -               | -             | -          |
| <b>PC5.</b> inform, escalate, or raise alarm about any suspicions, unaccounted hazardous material, devices, or other objects found in the premises                                                                                        | -            | -               | -             | -          |
| <b>PC6.</b> inform, escalate, or raise alarm about any breach of safety or security procedure in the organisation                                                                                                                         | -            | -               | -             | -          |
| <b>PC7.</b> help achieve zero accidents goals at work                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC8.</b> avoid damage to sensitive electronic components due to negligence of ESD procedures                                                                                                                                           | -            | -               | -             | -          |
| <b>PC9.</b> participate regularly in fire drills or other safety related workshops organised by the organisation                                                                                                                          | -            | -               | -             | -          |
| <b>PC10.</b> follow strictly all access control and perimeter safety procedures in designated factory areas such as robotic work stations, automated production lines, automated material movement and other potentially risky operations | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> ensure that other people follow all access control and perimeter safety procedures in designated factory areas and help avoid accidents                                  | -            | -               | -             | -          |
| <b>PC12.</b> use emergency switches or other mechanisms of stopping a machine immediately in case any emergency situation has developed or about to happen                            | -            | -               | -             | -          |
| <b>PC13.</b> ensure that electrical equipment are properly grounded                                                                                                                   | -            | -               | -             | -          |
| <b>PC14.</b> follow Cyber Security guidelines and be vigilant at workplace                                                                                                            | -            | -               | -             | -          |
| <b>PC15.</b> proceed to designated safe assembly area immediately on hearing fire alarm                                                                                               | -            | -               | -             | -          |
| <i>Follow healthy practices and posture</i>                                                                                                                                           | <b>8</b>     | <b>10</b>       | -             | -          |
| <b>PC16.</b> wash hands and use sanitizers as recommended to prevent spread of diseases                                                                                               | -            | -               | -             | -          |
| <b>PC17.</b> follow common personal hygiene practices                                                                                                                                 | -            | -               | -             | -          |
| <b>PC18.</b> maintain appropriate posture, especially in long hours of sitting or standing position and in handling heavy materials                                                   | -            | -               | -             | -          |
| <b>PC19.</b> participate in company organised health sessions such as exercises, games, yoga, physiotherapy, and other activities                                                     | -            | -               | -             | -          |
| <b>PC20.</b> handle heavy and hazardous materials with care, while maintaining appropriate posture, using suitable tools, and handling equipment such as trolleys, jacks, and ladders | -            | -               | -             | -          |
| <b>PC21.</b> learn and apply first aid devices available in the workplace                                                                                                             | -            | -               | -             | -          |
| <b>PC22.</b> learn and apply safety and handling procedures for electrical shock and electrocution                                                                                    | -            | -               | -             | -          |
| <b>PC23.</b> learn and apply emergency medical help services                                                                                                                          | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC24.</b> follow workplace decorum and avoid emotional outbursts or inappropriate language                                                                                                                             | -            | -               | -             | -          |
| <b>PC25.</b> prevent any harassment at workplace                                                                                                                                                                          | -            | -               | -             | -          |
| <i>Practice waste management and recycling</i>                                                                                                                                                                            | -            | -               | -             | -          |
| <b>PC26.</b> identify recyclable, non-recyclable, and hazardous waste generated in the workplace and comply with their disposal procedures                                                                                | -            | -               | -             | -          |
| <b>PC27.</b> dispose non-recyclable waste and hazardous waste following recommended processes                                                                                                                             | -            | -               | -             | -          |
| <b>PC28.</b> deposit recyclable and reusable material at identified locations                                                                                                                                             | -            | -               | -             | -          |
| <b>PC29.</b> support education and compliance of waste management processes                                                                                                                                               | -            | -               | -             | -          |
| <i>Conserve material and resources</i>                                                                                                                                                                                    | -            | -               | -             | -          |
| <b>PC30.</b> identify ways to optimize usage of material and resources such as water, electricity, energy in various tasks, activities, and processes                                                                     | -            | -               | -             | -          |
| <b>PC31.</b> check for spills and leakages of material in various tasks, activities, and processes and plug them                                                                                                          | -            | -               | -             | -          |
| <b>PC32.</b> escalate the leakage issue to appropriate authority if needed                                                                                                                                                | -            | -               | -             | -          |
| <b>PC33.</b> carry out routine cleaning of tools, machines, and equipment and maintain them in good working condition to optimize efficiency and wastage                                                                  | -            | -               | -             | -          |
| <b>PC34.</b> check if the equipment is functioning normally before commencing work and rectify or report any malfunctioning to the responsible agency                                                                     | -            | -               | -             | -          |
| <b>PC35.</b> check for any odour, sparks, fumes, emission, unusual vibration, noise, or any other objectionable presence in the environment and take immediate corrective action followed by report to responsible agency | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC36.</b> ensure electrical equipment are properly connected for use and are switched off when not in use | -            | -               | -             | -          |
| <b>PC37.</b> support education and compliance of resource conservation processes                             | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                             | <b>15</b>    | <b>20</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0505                                                                                                                                                                                                                                       |
| <b>NOS Name</b>            | Follow health, safety and environment guidelines at workplace                                                                                                                                                                                   |
| <b>Sector</b>              | Capital Goods                                                                                                                                                                                                                                   |
| <b>Sub-Sector</b>          | Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Defence Equipment, Fire-Fighting & Safety Equipment, Homeland Security |
| <b>Occupation</b>          | Service                                                                                                                                                                                                                                         |
| <b>NSQF Level</b>          | 5                                                                                                                                                                                                                                               |
| <b>Credits</b>             | 1                                                                                                                                                                                                                                               |
| <b>Version</b>             | 1.0                                                                                                                                                                                                                                             |
| <b>Last Reviewed Date</b>  | 01/10/2025                                                                                                                                                                                                                                      |
| <b>Next Review Date</b>    | 01/10/2030                                                                                                                                                                                                                                      |
| <b>NSQF Clearance Date</b> | 01/10/2025                                                                                                                                                                                                                                      |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings



## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 70% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

## Qualification Pack

**Minimum Aggregate Passing % at QP Level : 70**

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| CSC/N0914.Selection and Installation of Robotics & Automation Systems                        | 40           | 40              | 0             | 20         | 100         | 25         |
| CSC/N0915.Integration of Robots and Automation Systems using Industrial Networking Protocols | 40           | 40              | 0             | 20         | 100         | 25         |
| CSC/N0916.Testing & Maintenance of Robotics & Automation System.                             | 40           | 40              | 0             | 20         | 100         | 25         |
| CSC/N1339.Collaboratively coordinate with the team                                           | 30           | 70              | -             | -          | 100         | 10         |
| CSC/N0505.Follow health, safety and environment guidelines at workplace                      | 15           | 20              | -             | -          | 35          | 10         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                                | 20           | 30              | -             | -          | 50          | 5          |
| <b>Total</b>                                                                                 | <b>185</b>   | <b>240</b>      | <b>-</b>      | <b>60</b>  | <b>485</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |



## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |